

3. Material shared in village meetings for capacitor demonstration (Objective F)

3.1 Capacitor demonstration

Capacitor usage demonstration is a participatory process requiring the active engagement of various stakeholders: farmers, MSEDCL officials, and agriculture officials. The comprehensive methodology for demonstrating capacitor usage was given in the Phase III report of the MoUV. As part of the organized capacitor demonstration, the IIT Bombay energy team systematically conducted the following activities :

- **Identification of villages and DTs**
 - Villages identified through (i) secondary high-level data of the distribution of pumps, sprinklers, drip (ii) Krishi sahayak (iii) MSEDCL
 - Overloaded DTs identified through village meeting; MSEDCL DT failure data
- **Awareness creation**
 - Village-level meeting to introduce intervention
 - Measurement conducted on several pumps of a DT to show the effect of a single capacitor - voltage increase, current reduction
 - Awareness-building material circulated among farmers
- **Installation of capacitor**
 - Capacitor requirement collected at each DT (Number of pumps, HP; Capacitors purchased in bulk quantity; Installation)
- **Presentation of results in village meeting**
 - After the rabi season, a Village meeting to highlight the effect of capacitors for DT where the demonstration was done

Presently, village meetings are being conducted in selected village to demonstrate the effect of capacitors. Farmers on selected DTs are being surveyed to note down pump breakdown, present status of capacitor usage. The analysis of these farmers and summary of village meetings will be given in Phase V report.

3.2 Material shared in village meetings for capacitor demonstration

Capacitor demonstration is a yearlong exercise, which includes various activities as mentioned above. Throughout the capacitor demonstration, multiple village meetings were held. During the different stages, certain kinds of material were shared with farmers; so that they can get a good understanding of capacitor usage and its benefits. As a part of MoU V, during the process of capacitor demonstration following material was shared in village meetings:

1. Capacity-building material for key stakeholders in the workshop
2. The awareness-building material of capacitors on the irrigation pump
3. Capacitor installation guidelines: Scientific method of capacitor installation
4. Measurements taken on the pump before and after capacitor installation
5. Results of capacitor installation on selected DTs

3.2.1 Capacity building material for key stakeholders

As a part of phase II, a training workshop was conducted at IIT Bombay. During the workshop, stakeholders were educated on the intervention of capacitor usage on irrigation pumps. The capacity-building material is given in annexure C.

3.2.2 Awareness-building material for farmers

All important stakeholders should know about capacitors and ongoing interventions. Thus Capacitor awareness material needs to be circulated through different channels. The following kind of awareness-building material needs to be circulated among farmers. The templates of awareness-building material are given in annexure C.

1. Banners indicating ongoing interventions
2. Capacitor awareness messages to be spread through smartphones

Figure 3.1 Banner put in different locations (GP office, MSDCL premises) to spread the word regarding capacitor demonstration	Figure 3.2 Capacitor awareness messages spread through smartphones by JEs/AEs (to the WhatsApp groups of Zero Wiremen, Linemen and JEs along with a few selected villagers)

3.2.3 Readings taken during a demonstration of capacitor usage

As mentioned in the phase III report, taking measurement of readings on the pump in presence of farmers is an important aspect of capacitor demonstration. The format was finalised to note down the types of reading to be taken on pumps. The measurements were taken on all DTs as a part of the capacitor demonstration. The format for measurements of reading is given in annexure C. The following table represents a summary of the measurements taken:

Table: Measurements taken on DTs

		Before Capacitor			After Capacitor		
Village	Pump HP	Voltage (V)	Current (I)	PF	Voltage (V)	Current (I)	PF
Tigaon, Beed	3	418	9.8	0.87	429	8.7	0.96
Shivni, Beed	3	423	7.0	0.86	430	6.0	0.91
Rohana, Wardha	3	404	5.6	0.72	409	5	0.77
Junona, Wardha	3	399	6.2	0.5	403	5.3	0.6
Shivni, Beed	5	431	12.31	0.87	440	11.9	0.89
Kanholi, Wardha	5	409	12.2	0.82	412	10.1	0.85

3.2.4 Capacitor Installation Guidelines

Farmers' groups on selected DTs were created on WhatsApp. Informational videos regarding capacitors (Capacitor function, use and Benefits) and scientific methods of installing capacitors were shared with farmers through WhatsApp. The capacitor installation guideline and video link are given in Annexure C.

3.2.5 Results of capacitor installation on selected DTs

Farmers generally adopt new practices and interventions from nearby farmers. Providing evidence and testimonials to prove the effectiveness and reliability of a capacitor is an important step in the structured demonstration of capacitors. At the end of the Rabi season, village meetings have been organised to show the results of the capacitor. The template of the presentation to be given in the village meeting is given as a separate annexure C.

3.3 Present status of capacitor usage on selected DTs

Several DTs were selected for structured demonstration of the value of capacitors. The details of selected DTs for both Beed and Wardha districts were mentioned in the Phase III report. A capacitor demonstration was conducted in phase III of MoU V. After the capacitor demonstration capacitors were installed on pumps. Continuous guidance was provided to farmers during Rabi Hangam and also feedback was collected from them. The present status of capacitor usage is as follows:

Table 3.1. Details of selected DTs for Beed district

Sr. No.	Village	Name of the DT	kVA rating	No. of Pumps	Present capacitor usage
1	Bagpimpalgaon	Canal DT	100 kVA	18	15
		Umapur fata DT	63 kVA	14	14
2	Rui	Baman mala DT	63 kVA	21	14
3	Tigaon	Bag DT	100 kVA	31	31
4	Shivni	Kakarwadi DT	63 kVA	22	19
5	Pimpalwandi	Babu Shinde DT	100 kVA	13	13
6	Ekruka	Vashisht Kedar DT	100 kVA	31	25

Table 3.2. Details of selected DTs for Wardha district

Sr. No.	Village	Name of the DT	kVA rating	No. of Pumps	Present capacitor usage
1	Kanholi	Singhan DT	100 kVA	19	18
2	Rohana	Raut DT	100 kVA	19	17
3	Nandpur	Dondal DT	100 kVA	18	16
4	Junona	Giri DT	100 kVA	24	22